Trig 4

Review

1.
$$8^3 \times 6^{\frac{1}{2}} \div 32^{\frac{3}{2}} = \frac{512 \times \sqrt{6}}{\left(\sqrt{32}\right)^3}$$
$$= \frac{512\sqrt{6}}{4\sqrt{2}} \quad = 128\sqrt{3}$$

2. b

3.
$$\begin{aligned} 2x^2 + (2x-3)^2 &= 21 \\ 6x^2 - 12x + 9 &= 21 \\ 2x^2 - 4x - 4 &= 0 \end{aligned}$$

   So the correct answer is a

4. $f(x) = 6x^3 + 7x^2 - x - 2$

   $f(-1) = 0 \therefore (x+1)$ is a factor of $f(x)$

   Then by inspection:
$$\begin{aligned} f(x) &= (x+1)(6x^2 + x - 2) \\ &= (x+1)(3x+2)(2x-1) \end{aligned}$$

   So $x = -1$ or $x = -\frac{2}{3}$ or $x = \frac{1}{2}$.

5. Let $f'(x) = 3x^2 - 2x + 1$
$$\int 3x^2 - 2x + 1 dx = x^3 - x^2 + x + c$$

   We know that $f(2) = 5$.
$$\begin{aligned} f(2) &= 2^3 - 2^2 + 2 + c \\ &= 8 - 4 + 2 + c \quad = 6 + c \end{aligned}$$

   So $c = -1$.

Consolidation

1. i. $\frac{2}{3}\pi$
   ii. $\frac{2}{9}\pi$
   iii. $\frac{51}{36}\pi$

2. i. $1.2^c, 5.1^c$
   ii. $2.3^c, 5.4^c$
   iii. $1.6^c, 4.7^c$

3. i.
$$\begin{aligned} \frac{\sin x}{2\sin x \cos x} &= 1 \\ \frac{1}{2}\tan x &= 1 \\ \tan x &= 2x = 1.1^c, 4.2^c \end{aligned}$$

ii.
$$\begin{aligned}
2\sin^2 x - \cos x - 1 &= 0 \\
2\left(1-\cos^2 x\right) - \cos x - 1 &= 0 \\
-\cos^2 x - \cos x + 1 &= 0 \\
\cos x &= \frac{-1 \pm \sqrt{5}}{2} \\
-1 - \frac{\sqrt{5}}{2} &< 1 \\
\therefore \cos x &= \frac{-1+\sqrt{5}}{2} \\
x &= 0.904^c
\end{aligned}$$

4. i.
$$\begin{aligned}
2x &= -330°, -30°, 30°, 330° \\
x &= -165°, -15°, 15°, 165° \\
&= -\frac{11}{6}\pi, -\frac{1}{6}\pi, \frac{1}{6}\pi, \frac{11}{6}
\end{aligned}$$

ii.
$$\begin{aligned}
\tan x &= \frac{1}{\sqrt{3}} \\
x &= 30° \\
&= \frac{1}{12}\pi, -\frac{1}{12}\pi
\end{aligned}$$

iii.
$$\begin{aligned}
2\sin^2 x - \sin x - 1 &= 0 \\
(2\sin x + 1)(\sin x - 1) &= 0
\end{aligned}$$

So $\sin x = -\frac{1}{2}$ or $\sin x = 1$.
So $x = -\frac{5}{6}\pi, -\frac{1}{6}\pi$ or $= x = \frac{\pi}{2}$

iv.
$$4 - 5\sin x = 2\left(1-\sin^2 x\right) \sin^2 x - 5\sin x + 2 = 0$$

So $\sin x = \frac{5\pm\sqrt{17}}{2}$. As $\frac{5+\sqrt{17}}{2} > 1$, $x = \arcsin\left(\frac{5-\sqrt{17}}{2}\right) = 0.454$

5. i. $A = \pi \times r^2 = 100\pi = 314\text{cm}^2$

ii.
$$\begin{aligned}
A &= 6 \times \frac{10\sin 60° \times 10\cos 60°}{2} \\
&= 130\text{cm}^2
\end{aligned}$$

iii. Area of segments $= 314 - 130 = 184\text{cm}^2$

6.
$$\begin{aligned}
\frac{9.3}{10\pi} &= 0.296 \\
0.296 \times 360 &= 106.57°
\end{aligned}$$

Which is the angle formed by the two ends of the arc and the center of the circle. Then, the perpendicular distance of chord AB from the center of the circle can be found by

$$5\cos 53.285 = 2.99 \text{ cm}.$$

7.
$$\begin{aligned}\frac{d(2\pi r+2r)}{360}&=\frac{d\times\pi r^2}{360}\\ \left(\frac{d}{360}\right)(8\pi)+8&=\left(\frac{d}{360}\right)(16\pi)\\ \frac{8\pi d}{360}+8&=\frac{16\pi d}{360}\\ 8\pi d+2880&=16\pi d\\ 8\pi d&=2880\\ d&=114.6^\circ\end{aligned}$$

8. $r=2.5$m. The belt is perpendicular to the cylinder when it stops touching it, so the triangle formed has a right angle. Therefore,

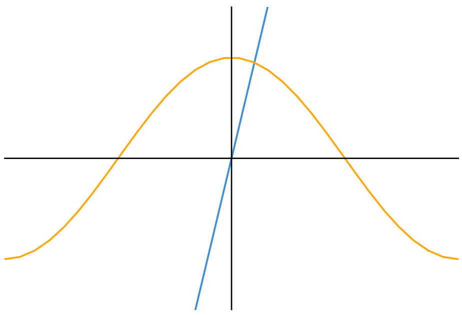
$$\sqrt{5^2-2.5^2}=4.334.33\times 2=8.66\text{ cm}$$

Which gives us the length of the belt not touching the cylinder. Then, to find the length of the belt touching the cylinder, we can find the angle formed by the two points where the belt touches the cylinder.

$$\begin{aligned}a^2&=b^2+c^2-2bc\cos A\\ 4.33^2&=5^2+2.5^2-2\times5\times2.5\times\cos A\\ \cos A&=-\frac{4.33^2-5^2-2.5^2}{2\times5\times2.5}\\ &=0.5A & &=60^\circ\end{aligned}$$

So the portion of the circumference that the belt touches is $\frac{240}{360}\times 2\times\pi 2.5=10.5$. So the total length of the belt is $8.66+10.5=19.1$m

9.



Exam Questions

1. a. area of sector $COD=\frac{1}{2}\times$ area of AOB.

$$\begin{aligned}2\times\frac{1}{2}\times|OC|^2\times0.8&=\frac{7\sin(0.8)\times10}{2}\\ |OC|&=\sqrt{\frac{7\sin(0.8)\times10}{2\times0.8}}\\ &=5.60\text{ cm}\end{aligned}$$

b. Perimeter is found by

$$\begin{aligned}&r\theta+(7-OD)+7+(10-OC)\\ &=5.6\times0.8+(7-5.6)+7+(10-5.6)\\ &=17.3\text{ cm}\end{aligned}$$

2. a.

$$\begin{cases} r\theta &= 15 \text{ cm} \\ \frac{1}{2}r^2\theta &= 45\text{cm}^2 \end{cases}$$

Therefore,

$$3r\theta = \frac{1}{2}r^2\theta$$
$$r = 6 \text{ cm}$$
$$\theta = 2.5^c$$

b. The area of the triangle formed by the two radii and the chord AB can be found by

$$\begin{aligned} h &= r\cos\frac{\theta}{2} \\ b &= 2r\sin\theta \\ A &= \frac{bh}{2} \\ &= \frac{r\cos\frac{\theta}{2} \times 2r\sin\frac{\theta}{2}}{2} \\ &= \frac{6\cos\frac{2.5}{2} \times 12\sin\frac{2.5}{2}}{2} \\ &= 10.27 \end{aligned}$$

The total area of the sector is $\frac{1}{2}r^2\theta = 45$ So the area of the segment bounded by the arc AB and the chord AB is $45 - 10.27 = 34.5\text{cm}^2$